



GIRIJANANDA CHOWDHURY UNIVERSITY

BCE311P	Solid Mechanics Laboratory	L	T	P	C
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Pre-requisite: Solid Mechanics, Engineering Materials and Application.					
Course Objectives: the objectives of this course are to:					
1. Basic knowledge in the field of Material Science. 2. Identify suitable Mechanical testing technique to inspect industrial component. 3. Apply knowledge of mathematics and engineering in calculating the mechanical properties of structural materials.					
Course Outcome: After successful completion of this course, the students will be able to					
CO 1: Contrasting the professional and ethical responsibility in the areas of material testing. CO 2: Develop the sense of techniques, skills and modern engineering tools necessary for engineering. CO 3: Examine and Evaluate the material Testing.					
Experiment No-1					4 hours
Tensile Test of a Ductile Material by using UTM					
Experiment No-2					4 hours
Compression Test of a Concrete cube by using UTM					
Experiment No-3					2 hours
Determination of impact strength by Charpy Impact Test					
Experiment No-4					2 hours
Determination of impact strength by Izod Impact Test					
Experiment No-5					4 hours
Determination of Hardness by Hardness Test					
Experiment No-6					4 hours
Determination of modulus of rigidity, torsional stiffness and ultimate share strength by Torsion Test					
Total Practical hours					20 hours
Text Book(s)					
1.	Material Testing Laboratory Manual For Quality Control, by Ravichawla Cb Kukreja,Kishore K (Author)				
2.	R. Subramanian, “Strength of Materials,” Oxford University Press, 2007.				
3.	D.S. Bedi, “Strength of Materials”, Khanna Book Publishing. 2022.				
4.	E. P. Popov, “Engineering Mechanics of Solids,” Pearson, 2015.				
5.	Composite Materials Design and Testing Hardcover – January 1, 2015.				
Reference Books					
1.	D.S. Bedi, “Engineering Mechanics”, Khanna Book Publishing. 2021				
2.	L.S Srinath, Advanced Mechanics of Solids, McGraw Hill, 2017				